

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CSE/ PEC-CS701C/ 2025-26
2025
NEURAL NETWORK AND DEEP LEARNING

Full Marks: 70

Times: 3 Hours

*The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.*

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer **all** questions

1. What is the difference between perceptron and McCulloch Pitts neuron?
2. What is flattening in a CNN model and why is it needed?
3. What is the effect of 3 x 3 convolution on a feature map?
4. Explain over-fitting in a CNN model?
5. Briefly describe sigmoid activation function.

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15 = 60

Answer any **four** questions

6. i) What is McCulloch-Pitts neuron? Show that a single McCulloch Pitts neuron can be used to represent any Boolean function which is linearly separable.
ii) Write some limitations of McCulloch Pitts neuron.
iii) How XOR problem is solved with two perceptrons?
7. i) Explain Perceptron learning algorithm with example.
ii) Given the following Boolean Function

x1	x2	Y
0	0	0
0	1	1
1	0	0
1	1	1

Implement this Boolean function using network of perceptrons with proper explanations.

8. i) Why Recurrent Neural Network (RNN) is needed?
ii) Explain RNN with suitable diagram and inputs.
iii) Expand the hidden layer of RNN and show how weight, bias and feedback connections of hidden layers are combined in forward propagation.
9. i) Show with diagram show sigmoid neuron is affected by change in values of weight W and bias b.
ii) What are the limitations of ANN? Using proper diagram explain robustness of a network.
iii) Describe overfitting and underfitting.
10. i) What is Regularization? Explain any three different regularization techniques.
ii) Describe how bias and variance affects overfitting and underfitting.
11. Derive the expressions for weight update
i) between the hidden and output layer.
ii) between the hidden and input layer of an MLP with single hidden layer.

P.T.O

1. ✓
- i) Describe vanishing gradient problem. How can ReLU overcome this problem?
 - ii) What is pooling layer. Give examples of different types of pooling operation.
 - iii) Find expression for output feature map size (Height, Width, Channel) in convolution operation.
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 2025
SIGNAL AND SYSTEMS

Full Marks: 70

Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer **all** questions

5x2=10

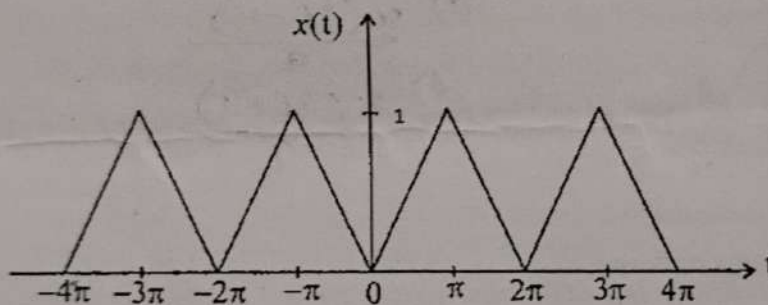
1. State time shifting property of Fourier series.
2. Write down the various steps of linear convolution.
3. Define Zero padding with example.
4. Prove that $r(t-2)=(t-2)u(t-2)$.
5. Draw $\text{rect}(t/4+0.25)$.

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any **five** questions

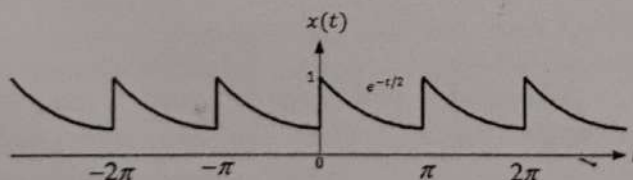
4x15=60

6. Determine the Exponential and Trigonometric Fourier series Expansion of the following signal. 15



1 -75.16
 2 -82.87
 3 -85.23

7. a) Discuss the Dirichlet Conditions with example. 5
 b) Discuss Causal and Noncausal system with example. 5
 c) Determine with examples (i) EVEN. ODD (ii) ODD. ODD 5
8. a) Find the continuous time Trigonometric Fourier series of the following signals and draw the line and phase spectrum 10



0 0.504
 11) 0.234
 12) 0.120

- b) Prove that the fourier series expansion of an Even periodic signal contains only cosine terms and a constant. 5
9. a) Find the continuous time Fourier Transforms of the following signals. 10
 - i. $f(t) = e^{-at}u(t)$
 - ii. $f(t) = u(t)$
 - iii. $f(t) = A \text{rect}(t/\tau)$
 - iv. $f(t) = \text{sgn}(t)$
- b) Discuss the advantages of Laplace transform. 5

10. a) Draw the Linear Convolution of the following with graphical method only.

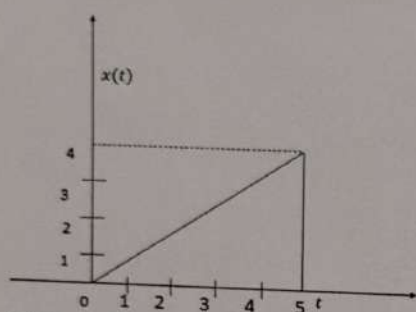
$$x(n) = \{1, 2, 2, 1\}; h(n) = \{1, 2, 2, 2, 1\}$$

b) Draw the circular convolution of the following using graphical method only.

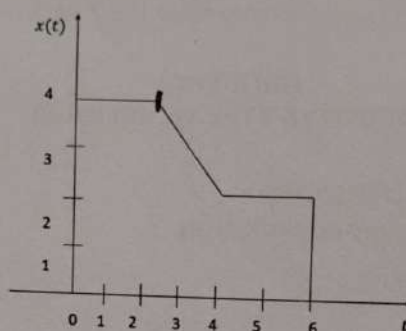
$$x_1(n) = \{1, 1, 2, 2\} \text{ and } x_2(n) = \{3, 3, 4, 4\}$$

11. a) Sketch and label the even and odd components of the signals shown below:

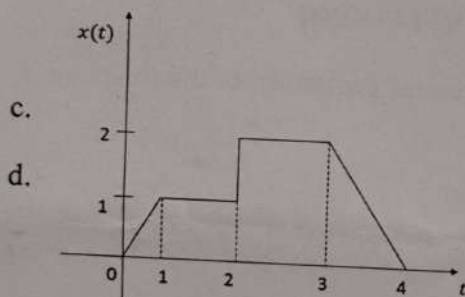
i)



ii.



b) Draw the following signals where $x(t)$ is given below:



c.

d.

a. $x(3t + 4)$

b. $x(3t - 4)$

c) $\frac{x(3t-4)}{9}$

d) $\frac{x(3t+4)}{9}$

c) Determine whether the signal is Energy or Power:

i. $x(t) = Ae^{j\omega t}$

ii. $x(t) = tu(t)$

First
Internal

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
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JGEC/B.TECH/CSE/PEC-CS702D/2025-26
2025
MACHINE LEARNING

Full Marks: 70

Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

5x2=10

1. State the applications of Machine Learning.
2. How the logistic regression can be used in prediction purpose?
3. Why elbow method is used in the K-Means algorithm?
4. What is the trade-off between bias and variance?
5. What is the difference between hard voting and soft voting?

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

4x15 = 60

6. i) Why Support Vector Machine (SVM) is an example of a maximum margin classifier? What are Support Vectors in SVMs? What are hard margin and soft margin SVMs? Why kernel trick is used in SVM? 3+2+2+3
ii) Explain the K-Means clustering algorithm with an example. 5
7. i) What is an Ensemble learning method? Compare the different types of ensemble learning methods? 2+3
ii) What is a Random Forest? How does it work? 2+3
iii) What are overfitting and underfitting in machine learning? How it can be addressed? 3+2
8. i) What is the difference between correlation and regression? Explain the different types of regression mechanisms. 2+3
ii) How is Principal Component Analysis (PCA) used for dimensionality reduction? How is the first principal component axis selected in PCA method? 5+5
9. i) Explain the K-NN algorithm with an example. Why it is called a lazy learner? 5+2
ii) Explain the Naive Bayes classifier with an example. What are the different types of Naive Bayes classifiers? 6+2
10. i) Explain the different learning paradigms that exist in machine learning with examples. 5
ii) Why is maximum likelihood estimation important in machine learning? Explain the mathematical background of the maximum likelihood estimation (MLE) algorithm. 2+8
11. i) Explain the different layers in CNN? Why do we use a Convolutional Layer in CNN? What are the different types of Pooling operations? Explain their characteristics. 3+2+2+3
ii) Compare and contrast classification and regression. 3
iii) Why is the kernel trick used in the K-means algorithm? 2
12. Write short notes on (Any **three**): 5+5+5
 - i) Statistical Learning Theory
 - ii) Deep Learning
 - iii) Gaussian Mixture Model
 - iv) Reinforcement Learning
 - v) Bayesian Learning

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JGEC/B.TECH/CSE/OEC-CS702C/2025-26

2025

Introduction to GIS and Remote Sensing

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.

Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A

[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

5x2=10

1. How can we convert an aerial photograph into a digital format?
2. What are the advantages of microwave remote sensing compared to the optical remote sensing?
3. Explain the future trends of remote sensing.
4. What do you mean by spectral signature?
5. What is dielectric constant?

GROUP-B

[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

4x15 = 60

6. i) Define remote sensing. What are the components of remote sensing? 2+5
ii) Define active and passive remote sensing. How the remote sensing can be applied in climate change? 2+3
iii) What are the challenges in remote sensing? 3
7. i) Explain in detail the various platforms used in remote sensing. 5
ii) Explain the most common types of sensors used in remote sensing. 5
iii) Compare and contrast the Push-broom and Whisk-broom scanners. 5
8. i) What is the special feature of Geospatial data? 2
ii) What is the electromagnetic spectrum? Explain the properties of different types of spectrum/rays present in the electromagnetic spectrum. 2+5
iii) Explain the types and significance of photogrammetry. 6
9. i) Explain the different components of GIS? Explain the applications of active microwave remote sensing? 5+5
ii) Explain the different types of microwave bands along with their corresponding frequencies and wavelengths used in microwave remote sensing? 5
10. i) What is aerial photography? Explain the characteristics of aerial photography. 2+6
ii) What factors influence the aerial photography? 7
11. i) What is the difference between high oblique and low oblique photograph? 2
ii) What do you understand by photogrammetry? Explain the photogrammetry process. 2+6
iii) Write down the applications of photogrammetry. 5
12. Write short notes on (Any **three**): 5+5+5
 - i) History of Remote Sensing
 - ii) Side-looking Radar
 - iii) Passive Microwave Remote Sensing
 - iv) Remote Sensing Image Classification
 - v) Spatial Operations of GIS

Multiple
Scenes
Day
Weather
Time of Day
Scale
Stereoscopy
Details
Scale
Time Cost
Remote
Image

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HUMAN RESOURCE DEVELOPMENT AND ORGANIZATIONAL BEHAVIOUR

Full Marks: 70

Times: 3 Hours

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Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions:

1. What is personality?
2. What is attitude?
3. What is job satisfaction?
4. Define conflict.
5. What is stress?

5×2=10
2
2
2
2
2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions:

6. Explain stress management strategies. What are the various external and internal causes of stress? What is organizational behaviour?
7. Write the determinants of personality. Write about types of personality. Define perception. What are the factors that can affect perception?
8. Explain motive, motivation, and motivating. Write a note on common errors while judging others. Write various factors that can affect decision-making.
9. Explain the ERG theory of motivation by Clayton Alderfer. What do you understand by satisfaction? Define leadership.
10. Write a note on the trait theory of leadership. What stress management steps can an organization take to minimize stress among employees?
11. What is functional and dysfunctional conflict? Write various types of causes of conflict. What is Herzberg's motivation-hygiene theory?

4×15=60
5+7+3
3+3+3+6
4+6+5
7+3+5
7+8
4+4+7

*Herzberg's
Two-Factor
Theory
Situational
Factors, bonus
Leadership
growth*

Challenging

*Time
Setting
Group
Size
Noise*